

A List of Device Functionalities in the Evaluation Benchmark

Device Name	#Funcs	List of Functions
Ikea Sensor	3	Turn on, turn off, information transmission
Tuya Switch	3	Turn on, turn off, timing
Mi Smart Desk Lamp Lite	4	Turn on, turn off, brightness control, color temperature control
Mijia Smart Plug 3	5	Turn on, turn off, information transmission, timing, count down
Yeelight LED Bulb	6	Turn on, turn off, brightness control, color temperature control, switch color, rhythmic mode
Aqara Smart Home Hub 2	7	Turn on, turn off, add device, delete device, device management, information transmission, update
Tuya Mini Gateway	7	Turn on, turn off, add device, delete device, device management, information transmission, update
EnOcean Gateway	7	Turn on, turn off, add device, delete device, device management, information transmission, update
KNX Floor Heating	7	Turn on, turn off, information transmission, control temperature, timing, count down, lock
Rheem Water Heater	7	Turn on, turn off, information transmission, control temperature, flow statistics, lock, timing
Media Air Purifier	7	Turn on, turn off, information transmission, timing, count down, switch mode, timing
Petkit Feeding	7	Turn on, turn off, information transmission, video streaming, audio streaming, feeding, feeding setting
Sony Soundbar	7	Turn on, turn off, control volume, switch mode, timing, access the internet
Anova Cooker	8	Turn on, turn off, information transmission, control temperature, timing, control water cycle, start cooking, stop cooking
Dyson Fan	8	Turn on, turn off, information transmission, control weed speed, control temperature, humidification, rotate, timing
Asus Router	10	Turn on, turn off, information transmission, device management, delete device, set network, management network, measure network speed, LED, update
Netgear Router	10	Turn on, turn off, information transmission, device management, delete device, set network, management network, measure network speed, LED, update
Xiaoai Speaker	11	Turn on, turn off, alarm, play music, lighting, control volume, access the internet, information transmission, connect other device, delete other device, update
Epson Printer	11	Turn on, turn off, information transmission, setting, document transmission, print, copy, scan, access the internet, hotspot, update
TOSHIBA AC	11	Turn on, turn off, information transmission, light on, light off, control temperature, control wind speed, control wind direction, switch mode, count down, timing
Media AC	11	Turn on, turn off, information transmission, light on, light off, control temperature, control wind speed, control wind direction, switch mode, count down, timing
Media AC	11	Turn on, turn off, information transmission, light on, light off, control temperature, control wind speed, control wind direction, switch mode, count down, timing
Philips TV	11	Turn on, turn off, information transmission, control volume, control brightness, switch video source, timing, control color temperature, casting, lock, access the internet
Apple TV	11	Turn on, turn off, information transmission, control volume, control brightness, switch video source, timing, control color temperature, casting, lock, access the internet
Dreame Vacuum	12	Turn on, turn off, information transmission, moving, switch mode, washing, timing, positioning, video streaming, audio streaming, access the internet, draw a map
Roomba Vacuum	12	Turn on, turn off, information transmission, moving, switch mode, washing, timing, positioning, video streaming, audio streaming, access the internet, draw a map
Mijia Robot Vacuum	12	Turn on, turn off, information transmission, moving, switch mode, washing, timing, positioning, video streaming, audio streaming, access the internet, draw a map
Ezviz C6c Camera	13	Turn on, turn off, horizontal rotation, vertical rotation, video streaming, zoom in, zoom out, play custom voice, audio streaming, screen shot, access the internet, information transmission, update
Dahua Camera	13	Turn on, turn off, horizontal rotate, vertical rotate, video streaming, zoom in, zoom out, play voice, audio streaming, take photos, access the internet, update
Imou Camera	13	Turn on, turn off, horizontal rotate, vertical rotate, video streaming, zoom in, zoom out, play voice, audio streaming, take photos, access the internet, update
Hikvision Camera	13	Turn on, turn off, horizontal rotate, vertical rotate, video streaming, zoom in, zoom out, play voice, audio streaming, take photos, access the internet, update